# Quasi-static model of the chinese paper yo-yo

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## 1 Geometric model

The system can be modelled as a constant-width rectangular ribbon of negligible thickness wrapped around a uniform-radius cylinder at a uniform pitch. One end of the ribbon is fixed in place to prevent rigid body motion, while the other ("active" end) is displaced in the $z$ direction (the axis of the cylinder). Refer to Figure 1 for a visual representation of the system. The system has the following geometric constants:

- $L$: the length of the unrolled rectangle.
- $W$: the width of the unrolled rectangle. Assume $L \gg W$.
- $\theta_L$: the total wind angle of the helix, which is fixed based on the boundary conditions (the two ends of the ribbon cannot rotate about the $z$-axis relative to one another, only translate along it, so the total number of turns is a fixed topological property of how the ribbon was wound and does not change as it is pulled). $\theta_L = 2\pi n$ where $n$ is the number of winds around the $z$ axis (not necessarily integer).

The state of the system is parametrized by the parameter $x$, which describes how far the active end has displaced in the $z$ direction relative to the base end.

### 1.1 Coordinate systems

Positions on the helix can be described with local surface $u, v$ coordinates, which describe the unrolled rectangle. The $u$ direction runs along the length of the unrolled rectangle, and the $v$ direction runs across its width. Alternatively, positions can be described in 3d space using cylindrical $r, \theta, z$ coordinates.

To simplify the geometry, we assume that the helix has a constant radius $r$ and a constant pitch angle $\phi$ (the angle the local $\hat{u}$ makes with the $r, \theta$ plane). Equivalently, the local orthonormal frame $(\hat{u}, \hat{v})$ tangent to the ribbon surface is related to the cylinder's own natural frame $(\hat{\theta}, \hat{z})$ by a rotation through the angle $\phi$:

$$\hat{u} = \cos\phi\,\hat{\theta} + \sin\phi\,\hat{z}, \qquad \hat{v} = -\sin\phi\,\hat{\theta} + \cos\phi\,\hat{z}. \tag{1}$$

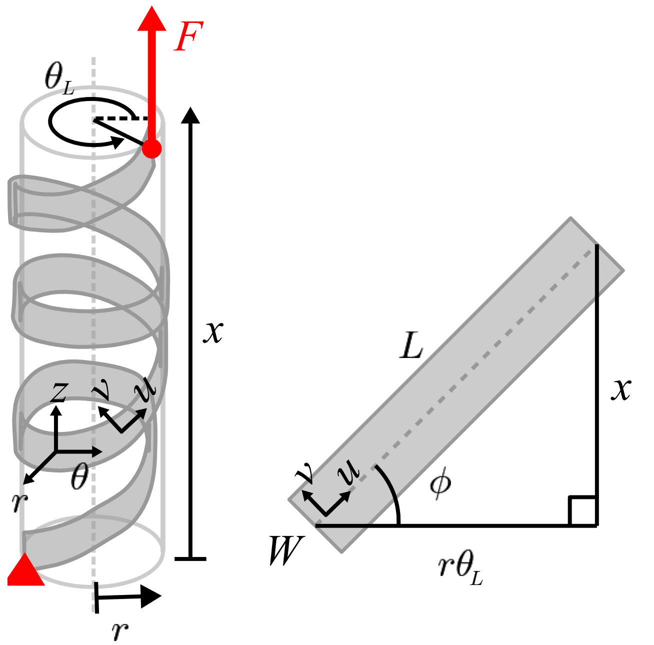


Figure 1: in progress

We now use Eq. (1) to relate the geometric constants $L, \theta_L$ to the pitch angle $\phi$ and radius $r$ at a given extension $x$. From Eq. (1), $\hat{u}$ has $z$-component $\sin\phi$, so moving a distance $\mathrm{d}u$ along the ribbon's own length produces an axial displacement $\mathrm{d}z = \sin\phi\,\mathrm{d}u$. Since $\phi$ is constant along the whole ribbon at a given (frozen) extension $x$, integrating this from the base end ($u = 0$) to the active end ($u = L$) gives a total axial displacement of $L\sin\phi$. By definition, this total axial displacement between the two ends of the ribbon is exactly $x$, so

$$x = L\sin\phi \qquad \Longrightarrow \qquad \phi = \sin^{-1}\left(\frac{x}{L}\right). \tag{2}$$

Likewise, $\hat{u}$ has $\theta$-component $\cos\phi$, so moving a distance $\mathrm{d}u$ along the ribbon produces a circumferential arc length $r\,\mathrm{d}\theta = \cos\phi\,\mathrm{d}u$. Integrating over the whole ribbon length $L$, the ribbon sweeps out a total circumferential arc length $L\cos\phi$. This total arc length must also equal $r\theta_L$ (arc length equals radius times swept angle), since by construction the ribbon winds through the fixed total angle $\theta_L$ regardless of $x$. Equating the two expressions for this arc length and using $\cos\phi = \sqrt{1-\sin^2\phi}$ together with Eq. (2) gives

$$r\theta_L = L\cos\phi \qquad \Longrightarrow \qquad r = \frac{L}{\theta_L}\cos\phi = \frac{L}{\theta_L}\sqrt{1-\frac{x^2}{L^2}} = \frac{\sqrt{L^2-x^2}}{\theta_L}. \tag{3}$$

### 1.2 Position vector

An area element $du \times dv$ at point $(u, v)$ has a position vector given in cylindrical coordinates. We build this up from Eq. (1): since $\hat{u} = \cos\phi\,\hat{\theta} + \sin\phi\,\hat{z}$, moving along the ribbon by $\mathrm{d}u$ produces $r\,\mathrm{d}\theta = \cos\phi\,\mathrm{d}u$ and $\mathrm{d}z = \sin\phi\,\mathrm{d}u$; since $\hat{v} = -\sin\phi\,\hat{\theta} + \cos\phi\,\hat{z}$, moving across the width by $\mathrm{d}v$ produces $r\,\mathrm{d}\theta = -\sin\phi\,\mathrm{d}v$ and $\mathrm{d}z = \cos\phi\,\mathrm{d}v$. Integrating these (at fixed, frozen $x$, so $r$ and $\phi$ are constants) from the reference point $(u,v)=(0,0)$ gives

$$\vec{p}(u,v,x)=\begin{bmatrix} r\\ \theta\\ z\end{bmatrix}=\begin{bmatrix} r\\ \frac{u\cos\phi-v\sin\phi}{r}\\ u\sin\phi+v\cos\phi\end{bmatrix}. \tag{4}$$

### 1.3 Local curvature

A cylinder of radius $r$ has two principal curvatures: $\kappa_\theta=1/r$ in the circumferential ($\hat{\theta}$) direction in which it is rolled, and $\kappa_z=0$ in the axial ($\hat{z}$) direction in which it is completely flat. Equivalently, the standard cylindrical basis vectors rotate with $\theta$ at the rates

$$\frac{\partial\hat{r}}{\partial\theta}=\hat{\theta},\qquad \frac{\partial\hat{\theta}}{\partial\theta}=-\hat{r},\qquad \frac{\partial\hat{z}}{\partial\theta}=0, \tag{5}$$

(the first two follow from writing $\hat{r}=\cos\theta\,\hat{x}+\sin\theta\,\hat{y}$ and $\hat{\theta}=-\sin\theta\,\hat{x}+\cos\theta\,\hat{y}$ in terms of the fixed Cartesian basis and differentiating; the third holds because $\hat{z}$ does not depend on $\theta$ at all — this is exactly the statement $\kappa_z=0$).

To differentiate the ribbon's own frame $(\hat{u},\hat{v},\hat{r})$ with respect to $(u,v)$ at a fixed $x$ (and therefore a fixed $r,\phi$), we take the partial derivatives of the $\theta$ component of Eq. (4):

$$\frac{\partial\theta}{\partial u}=\frac{\cos\phi}{r},\qquad \frac{\partial\theta}{\partial v}=-\frac{\sin\phi}{r}. \tag{6}$$

Combining Eq. (5) and Eq. (6) via the chain rule (e.g. $\frac{\partial\hat{r}}{\partial u}=\frac{\partial\hat{r}}{\partial\theta}\frac{\partial\theta}{\partial u}$) gives

$$\frac{\partial\hat{r}}{\partial u}=\frac{\cos\phi}{r}\hat{\theta},\qquad \frac{\partial\hat{r}}{\partial v}=-\frac{\sin\phi}{r}\hat{\theta},\qquad \frac{\partial\hat{\theta}}{\partial u}=-\frac{\cos\phi}{r}\hat{r},\qquad \frac{\partial\hat{\theta}}{\partial v}=\frac{\sin\phi}{r}\hat{r}, \tag{7}$$

with $\hat{z}$ constant with respect to $u,v$. Now we take the partial derivatives of $\hat{u},\hat{v}$ themselves, using Eq. (1) with $\phi$ held fixed:

$$\begin{aligned}\frac{\partial\hat{u}}{\partial u}&=\cos\phi\,\frac{\partial\hat{\theta}}{\partial u}=-\frac{\cos^2\phi}{r}\hat{r}, & \frac{\partial\hat{u}}{\partial v}&=\cos\phi\,\frac{\partial\hat{\theta}}{\partial v}=\frac{\sin\phi\cos\phi}{r}\hat{r},\\ \frac{\partial\hat{v}}{\partial u}&=-\sin\phi\,\frac{\partial\hat{\theta}}{\partial u}=\frac{\sin\phi\cos\phi}{r}\hat{r}, & \frac{\partial\hat{v}}{\partial v}&=-\sin\phi\,\frac{\partial\hat{\theta}}{\partial v}=-\frac{\sin^2\phi}{r}\hat{r}.\end{aligned} \tag{8}$$

The same three coefficients, $\cos^2\phi/r$, $\sin^2\phi/r$, and $\sin\phi\cos\phi/r$ (the last one twice), keep reappearing. We name them for convenience:

$$\kappa_{uu}:=\frac{\cos^2\phi}{r},\qquad \kappa_{vv}:=\frac{\sin^2\phi}{r},\qquad \kappa_{uv}:=\frac{\sin\phi\cos\phi}{r}, \tag{9}$$

Finally, we differentiate $\hat{r}$ itself. We already have $\frac{\partial\hat{r}}{\partial u},\frac{\partial\hat{r}}{\partial v}$ in terms of $\hat{\theta}$ (Eq. 7). To express them in terms of $\hat{u},\hat{v}$ instead, we invert Eq. (1) to obtain $\hat{\theta}=\cos\phi\,\hat{u}-\sin\phi\,\hat{v}$. Substituting this expression for $\hat{\theta}$ into Eq. (7) gives

$$\frac{\partial \hat{r}}{\partial u} = \frac{\cos\phi}{r}\big(\cos\phi\,\hat{u} - \sin\phi\,\hat{v}\big) = \kappa_{uu}\hat{u} - \kappa_{uv}\hat{v} \tag{10}$$

$$\frac{\partial \hat{r}}{\partial v} = -\frac{\sin\phi}{r}\big(\cos\phi\,\hat{u} - \sin\phi\,\hat{v}\big) = -\kappa_{uv}\hat{u} + \kappa_{vv}\hat{v} \tag{11}$$

Collecting everything derived in this section,

$$\begin{aligned}
\frac{\partial \hat{r}}{\partial u} &= \kappa_{uu}\hat{u} - \kappa_{uv}\hat{v}, & \frac{\partial \hat{r}}{\partial v} &= -\kappa_{uv}\hat{u} + \kappa_{vv}\hat{v},\\
\frac{\partial \hat{u}}{\partial u} &= -\kappa_{uu}\hat{r}, & \frac{\partial \hat{u}}{\partial v} &= \kappa_{uv}\hat{r},\\
\frac{\partial \hat{v}}{\partial u} &= \kappa_{uv}\hat{r}, & \frac{\partial \hat{v}}{\partial v} &= -\kappa_{vv}\hat{r}.
\end{aligned} \tag{12}$$

## 2 Stress balance

We will model the paper yoyo as a thin shell of small but finite thickness $t \ll r$. Because the ribbon's thickness ultimately carries the boundary tractions responsible for friction (Section 3), we work throughout with its full three-dimensional stress state. We use $\zeta \in [-t/2, t/2]$ for the coordinate measuring position through the thickness, i.e., displacement from the mid-surface along $\hat{r}$. At a point $(u, v, \zeta)$ the material carries the full Cauchy stress tensor:

$$\sigma = \begin{bmatrix} \sigma_{uu} & \sigma_{uv} & \sigma_{ur} \\ \sigma_{uv} & \sigma_{vv} & \sigma_{vr} \\ \sigma_{ur} & \sigma_{vr} & \sigma_{rr} \end{bmatrix} \tag{13}$$

There are six independent components: the in-plane stresses $\sigma_{uu}, \sigma_{uv}, \sigma_{vv}$; the transverse shears $\sigma_{ur}, \sigma_{vr}$ by which the ribbon can transmit a load sideways from one part of its width to another, and the normal pressure $\sigma_{rr}$ caused by layer-layer contact at the ribbon's surface.

We perform a force balance on a differential volume element $\mathrm{d}u \times \mathrm{d}v \times \mathrm{d}\zeta$. Each direction has a corresponding traction vector:

$$\vec{t}_u = \sigma_{uu}\hat{u} + \sigma_{uv}\hat{v} + \sigma_{ur}\hat{r} \qquad \vec{t}_v = \sigma_{uv}\hat{u} + \sigma_{vv}\hat{v} + \sigma_{vr}\hat{r} \qquad \vec{t}_\zeta = \sigma_{ur}\hat{u} + \sigma_{vr}\hat{v} + \sigma_{rr}\hat{r} \tag{14}$$

Because $t \ll r$, the element can be approximated as a rectangular prism, so the net force on the element is

$$\left(\frac{\partial \vec{t}_u}{\partial u} + \frac{\partial \vec{t}_v}{\partial v} + \frac{\partial \vec{t}_\zeta}{\partial \zeta}\right) \mathrm{d}u\,\mathrm{d}v\,\mathrm{d}\zeta = 0 \tag{15}$$

However, because $\hat{u}, \hat{v}, \hat{r}$ themselves depend on $u, v$ (Eq. 12), differentiating $\vec{t}_u, \vec{t}_v, \vec{t}_\zeta$ produces extra terms beyond the plain derivatives of the stress components. For example, expanding $\frac{\partial}{\partial u}(\sigma_{uu}\hat{u})$ introduces a curvature-weighted term in the $\hat{r}$ direction:

$$\frac{\partial}{\partial u}(\sigma_{uu}\hat{u}) = \frac{\partial \sigma_{uu}}{\partial u}\hat{u} + \sigma_{uu}\frac{\partial \hat{u}}{\partial u} = \frac{\partial \sigma_{uu}}{\partial u}\hat{u} - \kappa_{uu}\sigma_{uu}\hat{r} \tag{16}$$

The additional curvature-weighted terms reflect the fact that the element is not perfectly planar, so in-plane stresses are slightly rotated to have a small out-of-plane component, and vice versa. Expanding all terms and sorting by component gives three scalar force-balance equations:

$$\hat{u}: \quad \boxed{\frac{\partial \sigma_{uu}}{\partial u} + \frac{\partial \sigma_{uv}}{\partial v} + \frac{\partial \sigma_{ur}}{\partial \zeta} + \kappa_{uu}\sigma_{ur} - \kappa_{uv}\sigma_{vr} = 0} \tag{17}$$

$$\hat{v}: \quad \boxed{\frac{\partial \sigma_{uv}}{\partial u} + \frac{\partial \sigma_{vv}}{\partial v} + \frac{\partial \sigma_{vr}}{\partial \zeta} - \kappa_{uv}\sigma_{ur} + \kappa_{vv}\sigma_{vr} = 0} \tag{18}$$

$$\hat{r}: \quad \boxed{\frac{\partial \sigma_{ur}}{\partial u} + \frac{\partial \sigma_{vr}}{\partial v} + \frac{\partial \sigma_{rr}}{\partial \zeta} - \kappa_{uu}\sigma_{uu} + 2\kappa_{uv}\sigma_{uv} - \kappa_{vv}\sigma_{vv} = 0} \tag{19}$$

Equilibrium alone only gives 3 equations for 6 unknown stress components. However, because $t$ is small, we will make a first-order approximation that the stress field varies linearlly through $t$. This means that for any stress component, if we know its boundary value at both $-\frac{t}{2}, \frac{t}{2}$ for all $(u, v)$, that component is essentially solved. In the following section, we will obtain the surface boundary conditions for $\sigma_{ur}, \sigma_{vr}$, and $\sigma_{rr}$ based on layer-layer contact.

# 3 Friction

## 3.1 Overlapping neighbors

Because the ribbon is wound around itself, a given area element at $(u, v)$ could physically overlap with another area elements one full turn away along the ribbon. Specifically, an area element at point $(u, v)$ overlaps with the neighboring interior layer at point $(u^-, v^-)$ if their cylindrical coordinates have equal $r$ and $z$, but differ by one full turn in $\theta$. We can express this by taking the difference in $\vec{p}$:

$$\vec{p}(u, v, x) - \vec{p}(u^-, v^-, x) = \begin{bmatrix} 0 \\ 2\pi \\ 0 \end{bmatrix} \tag{20}$$

Plugging in Eq (4) gives:

$$u^- = u - 2\pi r \cos\phi, \tag{21}$$

$$v^- = v + 2\pi r \sin\phi. \tag{22}$$

By the identical argument with $\theta(u^+, v^+) = \theta(u, v) + 2\pi$ (i.e., the neighboring *outer* layer, one full turn further out), the element has an outer overlapping neighbor at

$$u^+ = u + 2\pi r \cos\phi, \tag{23}$$

$$v^+ = v - 2\pi r \sin\phi. \tag{24}$$

Some points $(u, v)$ on the surface overlap with an inner or outer layer, while others don't. We neglect the edge effects where a neighbor is absent because $u^-, u^+$ is outside the $[0, L]$ bounds. As a result, an inner neighbor is present exactly whenever $v^-$ lies within $[-W/2, W/2]$, and an outer neighbor is present exactly whenever $v^+$ lies in $[-W/2, W/2]$. That is,

$$\text{Inner neighbor is present if} \quad v \leq \frac{W}{2} - 2\pi r \sin\phi \tag{25}$$

$$\text{Outer neighbor is present if} \quad v \geq -\frac{W}{2} + 2\pi r \sin\phi \tag{26}$$

### 3.2 Friction direction and magnitude

The position vector $\vec{p}(u, v, x)$ of a material point $(u, v)$ is given by Eq. (4). Substituting Eq. (2) and Eq. (3) to expose the dependencies on $x$ gives:

$$\vec{p}(u, v, x) = \begin{bmatrix} \frac{\sqrt{L^2-x^2}}{\theta_L} \\ \frac{u\theta_L}{L} - \frac{v\,x\,\theta_L}{L\sqrt{L^2-x^2}} \\ \frac{ux}{L} + \frac{v\sqrt{L^2-x^2}}{L} \end{bmatrix} \tag{27}$$

The velocity of a given material point $(u, v)$ as the active end of the ribbon is displaced (i.e., as $x$ changes) is then obtained by taking the partial derivative of each component of Eq. (4) with respect to $x$:

$$\frac{\partial}{\partial x}\vec{p}(u, v, x) = \begin{bmatrix} \frac{\partial r}{\partial x} \\ \frac{\partial \theta}{\partial x} \\ \frac{\partial z}{\partial x} \end{bmatrix} = \begin{bmatrix} -\frac{x}{\theta_L\sqrt{L^2-x^2}} \\ -\frac{L\,\theta_L\,v}{(L^2-x^2)^{3/2}} \\ \frac{u}{L} - \frac{vx}{L\sqrt{L^2-x^2}} \end{bmatrix}. \tag{28}$$

We can then plug $u^-, v^-$ into Eq. (28) to compute the velocity of the element relative to its inner overlapping neighbor, which gives the relative slip direction in the $r, \theta, z$ frame:

$$\frac{\partial\vec{p}(u, v, x)}{\partial x} - \frac{\partial\vec{p}(u^-, v^-, x)}{\partial x} = \begin{bmatrix} 0 \\ \frac{2\pi\,x}{L^2-x^2} \\ \frac{2\pi}{\theta_L} \end{bmatrix}. \tag{29}$$

As expected, there is no relative velocity in the radial direction (i.e., the layers are not lifting away from each other). We can now compute the relative slip direction in the local frame by projecting this slip velocity from the $\theta, z$ basis onto the the $u, v$ basis using the relation described by Eq. (1). Since $\frac{\partial\theta}{\partial x}$ above is an *angular* rate, we first convert it to a linear (arc-length) rate by multiplying by $r$, giving $v_\theta = r\,\frac{\partial\theta}{\partial x}$; then projecting as just described:

$$\vec{v}_{slip}(u, v, x) = \begin{bmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} r\cdot\frac{2\pi\,x}{L^2-x^2} \\ \frac{2\pi}{\theta_L} \end{bmatrix} \tag{30}$$

We simplify this expression by substituting subsituting $x$ and $r$ in terms of $\phi$ (Eq. (2), (3)) to expose the double-angle combination, then applying double angle trig identities to factor out common terms. The result is a scaled unit vector:

$$\vec{v}_{slip}(u,v,x) = \frac{2\pi}{\theta_L \cos\phi}\begin{bmatrix}\sin(2\phi)\\ \cos(2\phi)\end{bmatrix} \tag{31}$$

The prefactor is strictly positive for all $x \in (0, L)$. Therefore, the relative slip velocity points in the direction of the unit vector $-(\sin 2\phi, \cos 2\phi)$ in the local $u, v$ frame, so the Coulomb friction force opposing this motion must point in the opposite direction:

$$\hat{f} = -\begin{bmatrix}\sin(2\phi)\\ \cos(2\phi)\end{bmatrix} \tag{32}$$

Recall that this is the friction exerted on the element by the inner layer, which resists the element from sliding forward with $x$. The components of this friction force are equivalent to the shear stresses acting on the inner $(-\hat{r})$ face. Additionally, the magnitude of the Coulomb friction force is equivalent to the normal pressure $-\sigma_{rr}$ times a friction coefficient $\mu$. By saying this, we assume $\sigma_{rr} \leq 0$ (the element is under compression), deferring a rigorous justification to Eq. (**??**) of Section 4. Therefore,

$$\sigma_{ur}\left(\zeta = -\frac{t}{2}\right) = -\mu\sigma_{rr}\left(\zeta = -\frac{t}{2}\right)\sin(2\phi) \tag{33}$$

$$\sigma_{vr}\left(\zeta = -\frac{t}{2}\right) = -\mu\sigma_{rr}\left(\zeta = -\frac{t}{2}\right)\cos(2\phi) \tag{34}$$

By the same argument, the friction exerted by the outer layer is the exact opposite, dragging the element forward. We then obtain a similar boundary relationship for the outer $(+\hat{r})$ face. Note that the sign is positive due to the sign convention for shear stresses on the inner face of an element.

$$\sigma_{ur}\left(\zeta = \frac{t}{2}\right) = -\mu\sigma_{rr}\left(\zeta = \frac{t}{2}\right)\sin(2\phi) \tag{35}$$

$$\sigma_{vr}\left(\zeta = \frac{t}{2}\right) = -\mu\sigma_{rr}\left(\zeta = \frac{t}{2}\right)\cos(2\phi) \tag{36}$$

### 3.3 Layer-layer contact relationship

By Newton's third law, two surfaces in contact should experience equal and opposite traction forces. Because the surface normals are opposite, this means they should have equal stresses. It suffices to equate the normal pressures $\sigma_{rr}$ because the transverse shear stresses are related above. Therefore, we get the following recursive relationships to define the boundary stresses, using piecewise gates based on layer contact boundaries:

$$\text{Outer surface:} \quad \sigma_{rr}\left(u, v, \zeta = \frac{t}{2}\right) = \begin{cases} \sigma_{rr}\left(u^+, v^+, \zeta = -\frac{t}{2}\right) & v \geq -\frac{W}{2} + 2\pi r \sin\phi \\ 0 & \text{otherwise} \end{cases} \tag{37}$$

$$\text{Inner surface:} \quad \sigma_{rr}\left(u, v, \zeta = -\frac{t}{2}\right) = \begin{cases} \sigma_{rr}\left(u^-, v^-, \zeta = \frac{t}{2}\right) & v \leq \frac{W}{2} - 2\pi r \sin\phi \\ 0 & \text{otherwise} \end{cases} \tag{38}$$

$$\tag{39}$$

The above relationships will recursively define the boundary stresses of the entire ribbon, using the boundary condition at the outermost layer $\sigma_{rr}(u = L) = 0$ as a base case.

# 4 PDE formulation and closed-form solution

We now assemble the pieces above into a closed system for the stress field. The three equilibrium equations, Eqs. (17)–(19), are three PDEs for six unknown stress components at every point $(u, v, \zeta)$. Section 3 supplies what is needed to close this at the two surfaces $\zeta = \pm\frac{t}{2}$: there, $\sigma_{ur}, \sigma_{vr}$ are fixed directly in terms of $\sigma_{rr}$ by Coulomb friction (Eqs. **??**–**??**), and $\sigma_{rr}$'s own two boundary values are tied recursively, layer to layer, by Newton's third law (Eqs. **??**–**??**). Combined with the linear-through-thickness approximation of Section 2, this reduces every unknown to a function of $(u, v)$ alone. We also carry forward the free-edge boundary conditions on the long, unloaded edges of the ribbon,

$$\boxed{\sigma_{uv}\left(u, v = \pm\frac{W}{2}\right) = 0, \qquad \sigma_{vv}\left(u, v = \pm\frac{W}{2}\right) = 0,} \tag{40}$$

which hold at every $\zeta$ and so apply directly to the thickness-uniform $\sigma_{uv}, \sigma_{vv}$ used below, and the end load applied at the active end, $F(x) = W\left[\sigma_{uu}(L)\sin\phi + \sigma_{uv}(L)\cos\phi\right]$ (the resultant of $\sigma_{uu}, \sigma_{uv}$ projected onto the global pulling direction).

## 4.1 Thickness integration

Integrate Eq. (17) over $\zeta \in [-\frac{t}{2}, \frac{t}{2}]$. Writing $\sigma_{ur}(u, v, \zeta)$ as linear in $\zeta$ with boundary values fixed by Eqs. (**??**),(**??**), its thickness average is

$$\bar{\sigma}_{ur} := \frac{1}{t}\int_{-\frac{t}{2}}^{\frac{t}{2}} \sigma_{ur}\, \mathrm{d}\zeta = \frac{1}{2}\Big[\sigma_{ur}(\frac{t}{2}) + \sigma_{ur}(-\frac{t}{2})\Big]$$

(the linear part of the profile integrates to zero over the symmetric interval), while $\int \frac{\partial \sigma_{ur}}{\partial \zeta}\, \mathrm{d}\zeta = \sigma_{ur}(\frac{t}{2}) - \sigma_{ur}(-\frac{t}{2}) =: f_u(u, v)$ is the same boundary-flux difference used previously. Taking $\sigma_{uu}, \sigma_{uv}$ uniform through the thickness (nothing at this order sources a $\zeta$-dependence in them) and dividing by $t$,

$$\boxed{\frac{\partial \sigma_{uu}}{\partial u} + \frac{\partial \sigma_{uv}}{\partial v} + \frac{f_u}{t} + \kappa_{uu}\bar{\sigma}_{ur} - \kappa_{uv}\bar{\sigma}_{vr} = 0,} \tag{41}$$

and, by the identical argument applied to Eq. (18),

$$\boxed{\frac{\partial \sigma_{uv}}{\partial u} + \frac{\partial \sigma_{vv}}{\partial v} + \frac{f_v}{t} - \kappa_{uv}\bar{\sigma}_{ur} + \kappa_{vv}\bar{\sigma}_{vr} = 0.} \tag{42}$$

Unlike a treatment that discards the curvature-weighted terms outright, Eqs. (41)–(42) keep them, now expressed through the through-thickness *average* $\bar{\sigma}_{ur}, \bar{\sigma}_{vr}$ rather than the pointwise value. Using Eqs. (**??**)–(**??**), $\bar{\sigma}_{ur} = -\frac{\mu}{2}\sin(2\phi)\big[\sigma_{rr}(\frac{t}{2}) + \sigma_{rr}(-\frac{t}{2})\big]$ and $\bar{\sigma}_{vr} = -\frac{\mu}{2}\cos(2\phi)\big[\sigma_{rr}(\frac{t}{2}) + \sigma_{rr}(-\frac{t}{2})\big]$ (each term vanishing on whichever side has no neighbor), and a short calculation from Eq. (9) shows $\kappa_{uu}\sin 2\phi - \kappa_{uv}\cos 2\phi = \kappa_{uv}$, so the curvature-weighted correction in Eq. (41) collapses to

$$\kappa_{uu}\bar{\sigma}_{ur} - \kappa_{uv}\bar{\sigma}_{vr} = -\frac{\mu\kappa_{uv}}{2}\Big[\sigma_{rr}(\frac{t}{2}) + \sigma_{rr}(-\frac{t}{2})\Big], \tag{43}$$

proportional to the *sum*, rather than the difference, of the two boundary pressures — and to $\kappa_{uv}$, not $\kappa_{uu}$. Because $\kappa_{uv}$ vanishes at $\phi = 0, 90^\circ$ and is exactly the off-diagonal curvature responsible for load transfer between $u$ and $v$ (Section 1.3), this term is a genuine, if secondary, correction to the friction-driven balance — not something to be dropped outright — and we carry it forward symbolically, returning to its relative size once $\sigma_{rr}$'s boundary values are pinned down below.

We now integrate the out-of-plane equation, Eq. (19), the same way. Its $\zeta$-derivative term integrates directly to the *jump* in $\sigma_{rr}$ across the thickness, $\Delta\sigma_{rr} := \sigma_{rr}(\frac{t}{2}) - \sigma_{rr}(-\frac{t}{2})$, while $\sigma_{uu}, \sigma_{uv}, \sigma_{vv}$ (uniform through the thickness) simply pick up a factor of $t$:

$$\boxed{\Delta\sigma_{rr} = t\big[\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv} + \kappa_{vv}\sigma_{vv}\big] - t\left[\frac{\partial \bar{\sigma}_{ur}}{\partial u} + \frac{\partial \bar{\sigma}_{vr}}{\partial v}\right],} \tag{44}$$

where the second bracket involves an extra in-plane derivative of an already boundary-traction-sized quantity and is subdominant to the first; we drop it below and return to it only if the leading balance proves insufficient. Equation (44) is the relation this section has been building toward: it says the two contact pressures felt on a ribbon element's two faces cannot be equal in general, but must differ by an amount set directly by the local in-plane tension, weighted by curvature and by the actual physical thickness $t$ — exactly the capstan-type mechanism (normal pressure $\sim$ tension over radius, for a tensioned band pressed against a curved surface) that generates contact pressure from tension in the first place. Writing out $\Delta\sigma_{rr} = \sigma_{rr}(\frac{t}{2}) - \sigma_{rr}(-\frac{t}{2})$ explicitly, Eq. (44) becomes

$$\boxed{\sigma_{rr}(u, v, \frac{t}{2}) - \sigma_{rr}(u, v, -\frac{t}{2}) \approx t\Big[\kappa_{uu}\sigma_{uu}(u, v) - 2\kappa_{uv}\sigma_{uv}(u, v) + \kappa_{vv}\sigma_{vv}(u, v)\Big].} \tag{45}$$

### 4.2 Closing the system: why $\sigma_{rr}$ does not simply propagate to zero

Read on its own, the matching condition of Section 3.3 (Eqs. **??**–**??**) contains no mechanism for generating a nonzero contact pressure anywhere: starting from the base case $\sigma_{rr}(L, v, \frac{t}{2}) = 0$, it would simply propagate that zero value backward through every layer, forcing $\sigma_{rr}(u, v, \pm\frac{t}{2}) \equiv 0$ everywhere, and hence, through Eqs. (**??**)–(**??**), $\sigma_{ur} \equiv \sigma_{vr} \equiv 0$ as well. It is Eq. (45) that resolves this: the matching condition and the pressure-jump relation hold *simultaneously*, not one after the other. Evaluating Eq. (45) at $(u^+, v^+)$ and substituting $\sigma_{rr}(u^+, v^+, -\frac{t}{2}) = \sigma_{rr}(u, v, \frac{t}{2})$ (Eq. **??**) gives

$$\boxed{\begin{aligned}\sigma_{rr}(u, v, \tfrac{t}{2}) = \sigma_{rr}(u^+, v^+, \tfrac{t}{2}) - t\Big[\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv} + \kappa_{vv}\sigma_{vv}\Big](u^+, v^+),\\ \text{valid for } v \geq -\tfrac{W}{2} + 2\pi r \sin\phi,\end{aligned}} \tag{46}$$

with $\sigma_{rr}(L, v, \frac{t}{2}) = 0$ starting the recursion. Unlike the matching condition alone, Eq. (46) carries a genuine source term at every step, $-t\big[\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv} + \kappa_{vv}\sigma_{vv}\big]$, nonzero whenever the local in-plane tension is nonzero. Unwound from the base case, $\sigma_{rr}(u, v, \frac{t}{2})$ is a sum of such terms evaluated at every wound layer further out than $(u, v)$ — so $\sigma_{rr}$ accumulates precisely because, and only because, $\sigma_{uu}$ (and $\sigma_{uv}, \sigma_{vv}$) are nonzero, exactly as anticipated. And $\sigma_{uu}$ is nonzero in turn because the Instron pull at the active end prescribes a nonzero resultant force $F(x)$ there (Section 4, opening): it is that end condition, not anything internal to Eqs. (41)–(45), that ultimately keeps every stress component in the system from relaxing to zero. (One can check directly that Eq. (**??**) combined with Eq. (45) at $(u, v)$ reproduces exactly Eq. (46) shifted by one layer, so the two contact relations of Section 3.3 are consistent with a single underlying recursion, not two independent ones.)

This also gives the complementary limit directly. If the helix is wound loosely enough that no $v$ satisfies either overlap condition at the current $x$ (Eq. **??**–**??**), then $\sigma_{rr}(u, v, \pm\frac{t}{2}) \equiv 0$ is forced pointwise *by definition* rather than by Eq. (46), and Eq. (45) then *requires* $\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv} + \kappa_{vv}\sigma_{vv} \equiv 0$ as a consistency condition. Together with $f_u = f_v = 0$ in Eqs. (41)–(42) and the free-edge conditions of Eq. (40), this drives $\sigma_{uu}, \sigma_{uv}, \sigma_{vv} \equiv 0$ as well: with no self-contact anywhere there is no mechanism to resist the pull, so the ribbon offers no static tension and simply slides freely. This matches the physical expectation exactly.

### 4.3 The accumulated contact pressure in closed form

Equation (46) steps $(u, v) \to (u^+, v^+) = (u + 2\pi r \cos\phi,\ v - 2\pi r \sin\phi)$ at each iteration — a fixed shift, since $r, \phi$ are frozen at any given extension $x$. Because the source term on the right of Eq. (46) carries an explicit factor of $t$, the change in $\sigma_{rr}(\cdot, \frac{t}{2})$ from one wound layer to the next is small (order $\frac{t}{r}$ relative to $\sigma_{uu}$ itself), even though the geometric step $2\pi r \cos\phi$ between layers is not. This is exactly the situation in which a discrete recursion is well approximated by its continuum limit: treating $\sigma_{rr}(u, v, \frac{t}{2})$ as a smooth function and expanding Eq. (46) to first order in the step gives the transport equation

$$2\pi r \cos\phi \, \frac{\partial \sigma_{rr}(u, v, \frac{t}{2})}{\partial u} - 2\pi r \sin\phi \, \frac{\partial \sigma_{rr}(u, v, \frac{t}{2})}{\partial v} = t\Big[\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv} + \kappa_{vv}\sigma_{vv}\Big]. \tag{47}$$

The left side is exactly $2\pi r \cos\phi$ times the derivative of $\sigma_{rr}(\cdot, \frac{t}{2})$ along the direction $(\cos\phi, -\sin\phi)$ — precisely the direction of the neighbor shift $(u^+, v^+)$ itself, per Eq. (23)–(24). If, as adopted below (Section 4.4), $\sigma_{uu}, \sigma_{uv}, \sigma_{vv}$ are taken independent of $v$, the right side of Eq. (47) depends only on $u$, and integrating Eq. (47) along its characteristic line from $(u, v)$ out to the base case at

$u' = L$ gives the closed form

$$\boxed{\sigma_{rr}(u, v, \frac{t}{2}) = -\frac{t}{2\pi r \cos\phi} \int_u^L \Big[\kappa_{uu}\sigma_{uu}(u') - 2\kappa_{uv}\sigma_{uv}(u') + \kappa_{vv}\sigma_{vv}(u')\Big] \,\mathrm{d}u',} \tag{48}$$

with $\sigma_{rr}(u, v, -\frac{t}{2})$ recovered directly from Eq. (45). Notably, Eq. (48) is then independent of $v$ as well, consistent with the $v$-independence assumed for $\sigma_{uu}, \sigma_{uv}, \sigma_{vv}$ on the right. This closed form implicitly assumes the characteristic through $(u, v)$ reaches $u' = L$ before its $v$-coordinate drifts outside $\left[-\frac{W}{2}, \frac{W}{2}\right]$ and exits through the side of the ribbon rather than its end; resolving that width-edge truncation exactly would replace the upper limit $L$ with $\min\Big(L,\ u + \left(v + \frac{W}{2}\right)\frac{\cos\phi}{\sin\phi}\Big)$, a secondary correction we do not carry through below.

### 4.4 Narrow-ribbon reduction and the closed ODE

As in a beam or narrow-plate approximation, and consistent with $L \gg W$ already assumed throughout, we take $\sigma_{vv}(u, v) \approx 0$ and $\sigma_{uu}, \sigma_{uv}$ approximately independent of $v$ (uniform across the width) for the remainder of this section; unlike the compatibility-based argument used in an earlier pass on this document, this is now adopted directly as a modeling approximation for a long, narrow ribbon, not derived. Integrating Eq. (41) over $v \in \left[-\frac{W}{2}, \frac{W}{2}\right]$: the $\frac{\partial \sigma_{uu}}{\partial u}$ term gives $W \frac{\mathrm{d}\sigma_{uu}}{\mathrm{d}u}$ (an ordinary derivative, since $\sigma_{uu}$ is now a function of $u$ alone); the $\frac{\partial \sigma_{uv}}{\partial v}$ term integrates, via the free-edge condition of Eq. (40), to $\sigma_{uv}\left(u, \frac{W}{2}\right) - \sigma_{uv}\left(u, -\frac{W}{2}\right) = 0$; and the $\frac{f_u}{t}$ term integrates using Eq. (**??**),(**??**) and the fact that $\sigma_{rr}\left(u, \cdot, \pm\frac{t}{2}\right)$ is itself $v$-independent (Eq. 48), so only the presence/absence gates of Eq. (**??**)–(**??**) depend on $v$, each contributing a factor equal to the length of its band, $W - 2\pi r \sin\phi$:

$$\begin{aligned} \frac{1}{tW} \int_{-\frac{W}{2}}^{\frac{W}{2}} f_u \,\mathrm{d}v &= -\frac{\mu \sin(2\phi)\,(W - 2\pi r \sin\phi)}{tW} \left[\sigma_{rr}\left(u, \frac{t}{2}\right) - \sigma_{rr}\left(u, -\frac{t}{2}\right)\right] \\ &= -\frac{\mu(W - 2\pi r \sin\phi)}{W} \sin(2\phi) \Big[\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv}\Big], \end{aligned} \tag{49}$$

using Eq. (45) (with $\sigma_{vv} \approx 0$) for the bracketed jump. The curvature-correction term of Eq. (43) instead needs the *sum* $\sigma_{rr}\left(u, \frac{t}{2}\right) + \sigma_{rr}\left(u, -\frac{t}{2}\right)$, not the difference — and by Eq. (48) that sum is set by the pressure *accumulated over every layer out to* $u = L$, not by the local tension at $u$ alone. Whether this nonlocal term is actually small compared to Eq. (49) therefore depends on how many layers are wound outside $u$ (it is not guaranteed small just because $t \ll r$); we drop it here to obtain a closed, local system, flagging it — honestly, not by assumption — as an open refinement, exactly as with the term already dropped in Eq. (44). Applying the identical steps to Eq. (42) gives the pair

$$\boxed{\begin{aligned} \frac{\mathrm{d}\sigma_{uu}}{\mathrm{d}u} &= \frac{\mu(W - 2\pi r \sin\phi)}{W} \sin(2\phi) \Big[\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv}\Big], \\ \frac{\mathrm{d}\sigma_{uv}}{\mathrm{d}u} &= \frac{\mu(W - 2\pi r \sin\phi)}{W} \cos(2\phi) \Big[\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv}\Big]. \end{aligned}} \tag{50}$$

(For $2\pi r\sin\phi > W$ — full overlap, no bare band at all — the coefficient $W - 2\pi r\sin\phi$ should be read as 0, not negative; we work in the regime $2\pi r\sin\phi < W$ below.)

### 4.5 Rank-one structure and closed-form solution

The right side of both equations in Eq. (50) is the same scalar, $\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv}$, multiplying $\sin(2\phi)$ in the first equation and $\cos(2\phi)$ in the second — so the $2\times 2$ coefficient matrix is rank one, with $(\sin 2\phi, \cos 2\phi)$ its only nonzero-eigenvalue eigenvector. Writing $\sigma_{uu}(u) = A(u)\sin(2\phi)$, $\sigma_{uv}(u) = A(u)\cos(2\phi)$ (the general solution also admits a null-space term along $(2\sin\phi, \cos\phi)$, which we set to zero, as before, for the minimal, best-conditioned solution carrying a single free constant), the bracket becomes

$$\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv} = A(u)\big[\kappa_{uu}\sin 2\phi - 2\kappa_{uv}\cos 2\phi\big] = A(u)\,\frac{2\sin^3\phi\cos\phi}{r}$$

(using $\kappa_{uu} = \frac{\cos^2\phi}{r}$, $\kappa_{uv} = \frac{\sin\phi\cos\phi}{r}$ from Eq. 9 and standard double-angle identities), and both equations of Eq. (50) collapse to the single scalar equation

$$\boxed{\frac{\mathrm{d}A}{\mathrm{d}u} = \frac{2\mu(W - 2\pi r\sin\phi)\sin^3\phi\cos\phi}{rW}\,A(u).} \tag{51}$$

Since $r, \phi$ are frozen at fixed $x$, the coefficient on the right is a constant in $u$, and Eq. (51) integrates immediately to

$$\boxed{A(u) = A_0 \exp\left[\frac{2\mu(W - 2\pi r\sin\phi)\sin^3\phi\cos\phi}{rW}\,u\right],} \tag{52}$$

with $A_0 := A(0)$ the single free constant carrying the physical scale of the whole solution (the holding tension at the fixed, base end), so that

$$\boxed{\begin{aligned} \sigma_{uu}(u) &= A_0\sin(2\phi)\exp\left[\frac{2\mu(W - 2\pi r\sin\phi)\sin^3\phi\cos\phi}{rW}\,u\right], \\ \sigma_{uv}(u) &= A_0\cos(2\phi)\exp\left[\frac{2\mu(W - 2\pi r\sin\phi)\sin^3\phi\cos\phi}{rW}\,u\right]. \end{aligned}} \tag{53}$$

### 4.6 Sign and monotonicity of the contact pressure

Section 3.2 assumed $\sigma_{rr} \le 0$ (compression) throughout, in order to write the Coulomb friction magnitude as $\mu$ times $-\sigma_{rr}$; we now verify this self-consistently. Take $A_0 > 0$ (a positive base tension). Since $\phi \in (0°, 90°)$, $\sin\phi, \cos\phi > 0$, so every prefactor appearing above is non-negative, and $\sigma_{uu}(u) = A_0\sin(2\phi)e^{(\cdots)u} \ge 0$ for all $u \in [0, L]$: the axial stress is tensile everywhere, as it must be for a ribbon that is only ever pulled, never pushed. Substituting Eq. (53) into the bracket of Eq. (48),

$$\kappa_{uu}\sigma_{uu}(u') - 2\kappa_{uv}\sigma_{uv}(u') = A_0\,\frac{2\sin^3\phi\cos\phi}{r}\,\exp\left[\frac{2\mu(W-2\pi r\sin\phi)\sin^3\phi\cos\phi}{rW}\,u'\right] \\ \geq 0 \quad \text{for all } u' \in [0, L], \tag{54}$$

so the integrand in Eq. (48) is non-negative throughout, its integral over $u' \in [u, L]$ is non-negative, and the leading minus sign in Eq. (48) gives

$$\sigma_{rr}\left(u, v, \frac{t}{2}\right) \leq 0 \quad \text{for all } (u, v), \tag{55}$$

confirming the compression assumed throughout Section 3. The same holds for $\sigma_{rr}\big(u, v, -\frac{t}{2}\big)$, since

$$\sigma_{rr}\left(u, v, -\frac{t}{2}\right) = \sigma_{rr}\left(u, v, \frac{t}{2}\right) - t\Big[\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv}\Big](u) \tag{56}$$

(Eq. 45) subtracts a term that is itself non-negative. Beyond the sign, Eq. (48) is manifestly *monotonic*: as $u$ decreases from $L$ (where it is exactly zero, by the base case) toward 0, the integration range $[u, L]$ only grows, so $\left|\sigma_{rr}\left(u, v, \frac{t}{2}\right)\right|$ increases monotonically. This matches the physical picture directly: a layer near the base end ($u = 0$) has every other wound layer stacked outside it and so carries the full accumulated contact pressure, while a layer at the active end ($u = L$) has nothing wound outside it and carries none.

## 4.7 Force-displacement relation

The resultant force at the active end, projected onto the global pulling direction, is $F(x) = W\big[\sigma_{uu}(L)\sin\phi + \sigma_{uv}(L)\cos\phi\big]$ (Section 4, opening). Substituting Eq. (53) at $u = L$ and using $\sin(2\phi)\sin\phi + \cos(2\phi)\cos\phi = \cos(2\phi - \phi) = \cos\phi$ to collapse the double-angle dependence,

$$F(x) = WA_0\cos\phi\,\exp\left[\frac{2\mu(W-2\pi r\sin\phi)\sin^3\phi\cos\phi}{rW}\,L\right]. \tag{57}$$

Writing this out in terms of $x$ via $\sin\phi = \frac{x}{L}$, $\cos\phi = \frac{\sqrt{L^2-x^2}}{L}$, $r = \frac{\sqrt{L^2-x^2}}{\theta_L}$ (Eq. 2–3), the exponent simplifies (the two factors of $\sqrt{L^2-x^2}$, one from $\cos\phi$ and one from $\frac{1}{r}$, cancel) to

$$\frac{2\mu(W-2\pi r\sin\phi)\sin^3\phi\cos\phi}{rW}\,L = \mu\left(1 - \frac{2\pi x\sqrt{L^2-x^2}}{\theta_L LW}\right)\frac{2\theta_L x^3}{L^3}, \tag{58}$$

giving the closed-form force–displacement law

$$\boxed{F(x) = \frac{WA_0\sqrt{L^2-x^2}}{L}\,\exp\left[\mu\left(1 - \frac{2\pi x\sqrt{L^2-x^2}}{\theta_L LW}\right)\frac{2\theta_L x^3}{L^3}\right],} \tag{59}$$

valid for $2\pi x\sqrt{L^2 - x^2} < \theta_L LW$ (Section 4.4), i.e., before the overlap band consumes the full width $W$. As in the leading-order picture of Section 4.2, $F(0) = \frac{WA_0}{L} \cdot L = WA_0$ is set entirely by the base tension $A_0$, and the exponential growth in $x$ is exactly the accumulated, capstan-type effect of Section 4.3: each additional bit of wound-over contact multiplies the required pulling force rather than adding to it.

# 5 Experimental validation

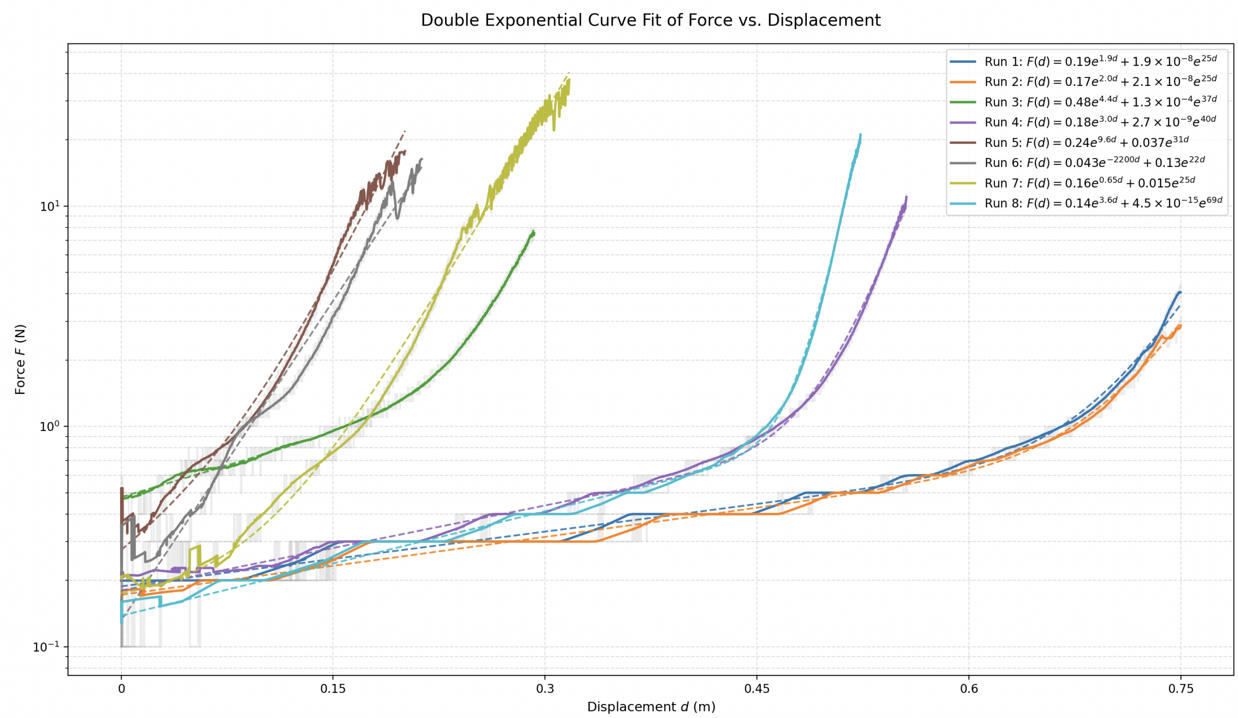


Figure 2: Instron force-displacement data, with raw quantized data (grey), smoothed data (solid colored), and curve fit (dashed colored).